Mextropic

Assay Capabilities

ASSAY CATALOGUE

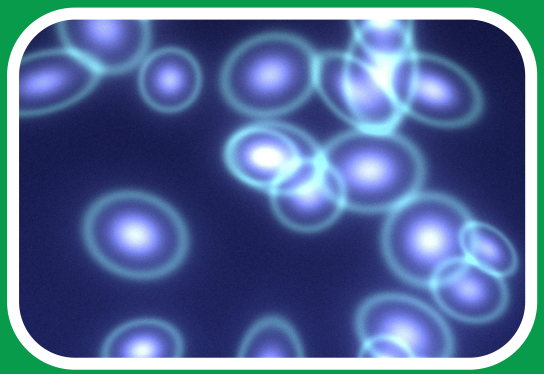
102 CAPABILITIES / 19 CATEGORIES

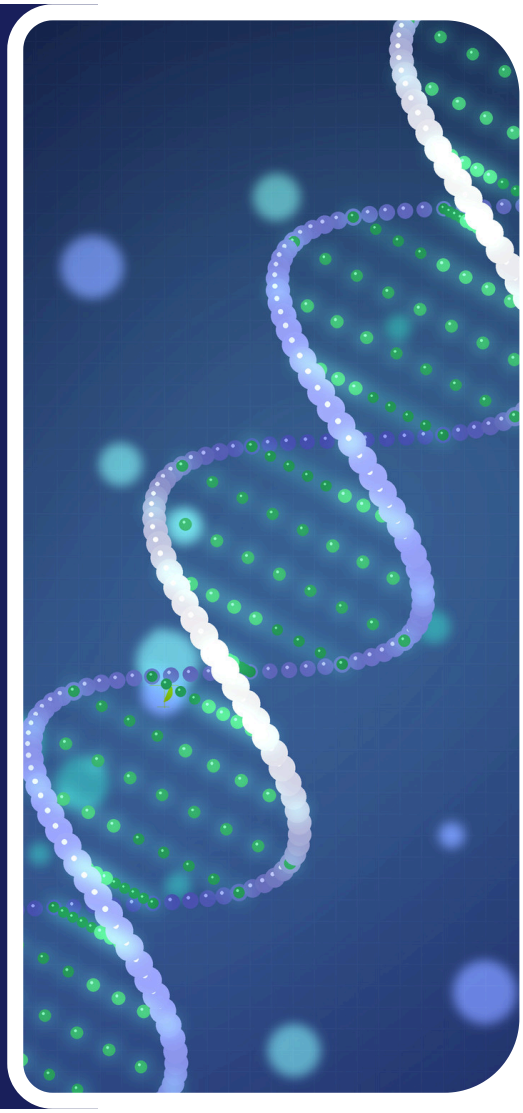
102

CAPABILITIES

19

CATEGORIES





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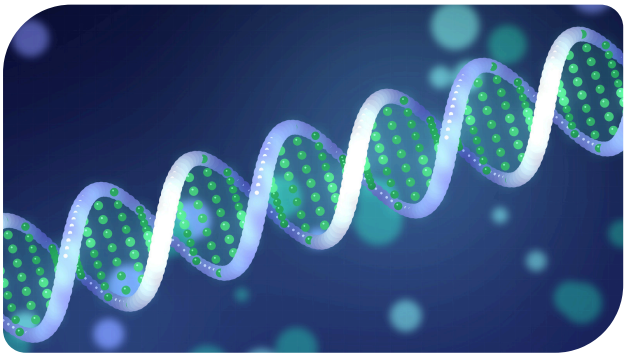
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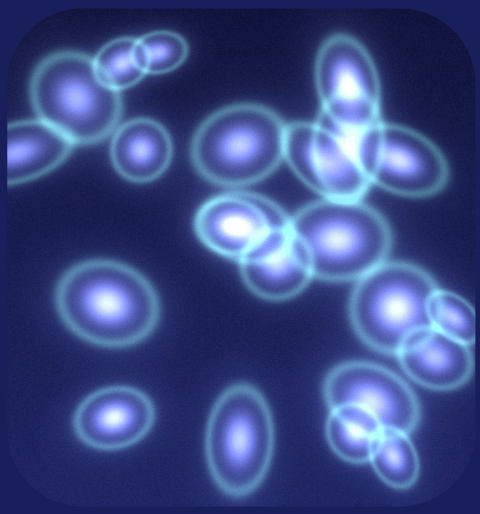
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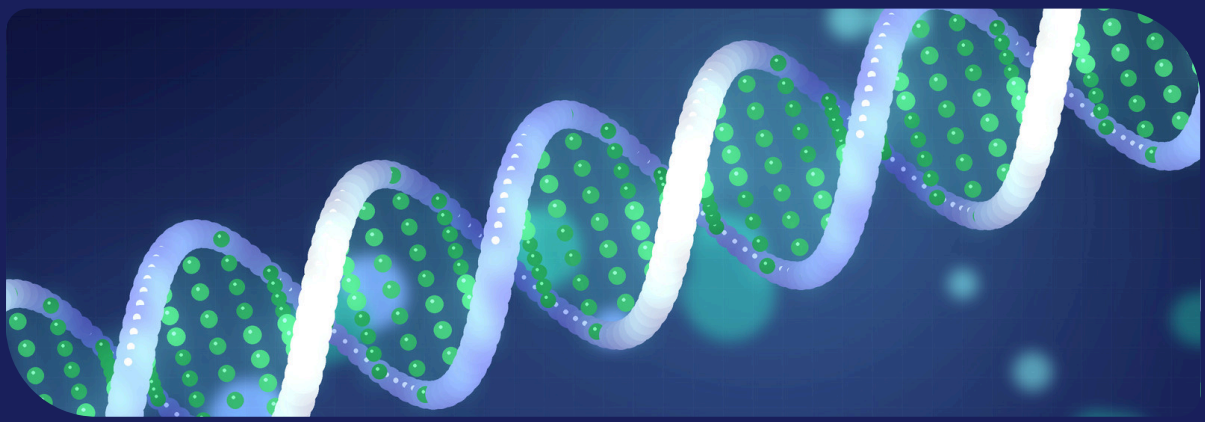


SECTION 01

Imaging and omics datasets

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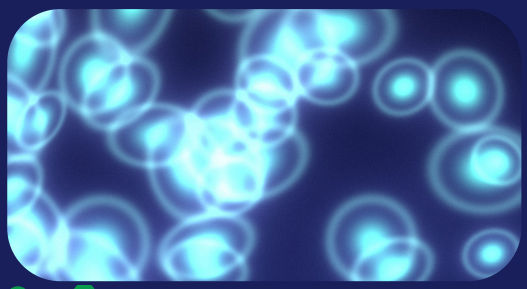




CATEGORY 01

Imaging

4 CAPABILITIES



High-content imaging

High-content imaging with multiplexed, image-based feature extraction on high-content screening platforms

Spatial transcriptomics

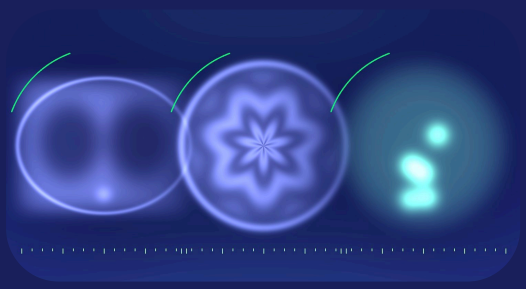
Spatially resolved transcript profiling of tissue sections, retaining the tissue coordinates of gene expression

Spatial proteomics (imaging MS)

Mass-spectrometry imaging of tissue sections, producing spatially resolved protein and peptide distribution maps

Histopathology and tissue staining

Histological preparation, microscopic assessment, and selected IHC/ICC workflows



CATEGORY 02

Clinical Imaging

20 CAPABILITIES

NOTE *Availability, delivery route, and the ethics and consent basis for human imaging data are confirmed per programme.*

MRI – structural

T1-, T2- and FLAIR-weighted magnetic resonance series covering soft-tissue and organ anatomy, delivered as DICOM

MRI – diffusion-weighted and DTI

Diffusion-weighted imaging and diffusion-tensor acquisitions with derived ADC and tractography maps

MRI - functional (BOLD)

Blood-oxygen-level-dependent functional MRI, task-based and resting-state

MRI - spectroscopy (MRS)

In vivo magnetic resonance spectroscopy for regional metabolite concentrations

Cardiac MRI

Cine, perfusion and late-gadolinium-enhancement cardiac magnetic resonance for function, viability and fibrosis

CT - non-contrast and contrast-enhanced

Multi-detector computed tomography series, with and without iodinated contrast

CT angiography

ECG-synchronised contrast-enhanced CT of coronary and peripheral vasculature

Cone-beam CT

Cone-beam computed tomography acquired in interventional and image-guided settings

Digital radiography (X-ray)

Planar digital projection radiography across chest, skeletal and abdominal indications

Fluoroscopy and interventional angiography

Real-time fluoroscopic and catheter-angiographic series from diagnostic and interventional procedures

Mammography and digital breast tomosynthesis

Full-field digital mammography and tomosynthesis volumes

Dual-energy X-ray absorptiometry (DEXA)

Bone mineral density and body-composition scans

Ultrasound - B-mode and Doppler

Greyscale and colour/spectral Doppler ultrasound across abdominal, vascular and obstetric indications

Echocardiography

Transthoracic two-dimensional, three-dimensional and strain echocardiography

Ultrasound elastography

Shear-wave and strain elastography for tissue stiffness, including hepatic fibrosis staging

PET/CT

Positron-emission tomography with co-registered CT, including FDG and non-FDG tracers

PET/MRI

Simultaneous positron-emission tomography and magnetic resonance acquisition

SPECT and SPECT/CT

Single-photon-emission computed tomography, with co-registered CT where acquired

Retinal imaging - fundus and OCT

Colour fundus photography and optical coherence tomography of the retina

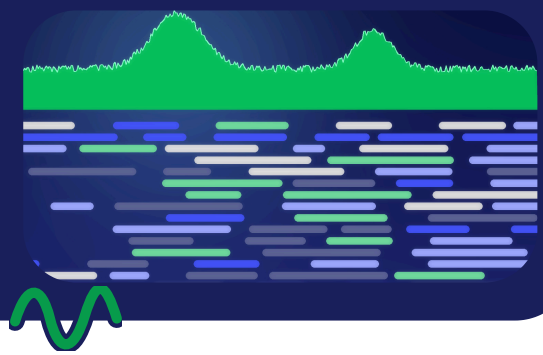
Digital pathology / whole-slide imaging

Scanned whole-slide H&E and immunohistochemistry images for quantitative review

CATEGORY 03

Genomics / Transcriptomics

5 CAPABILITIES



Bulk RNA-seq

Bulk RNA sequencing on current-generation Illumina short-read platforms

Single-cell / single-nucleus RNA-seq

Single-cell and single-nucleus RNA sequencing on Chromium-based platforms

Long-read / full-length isoform sequencing

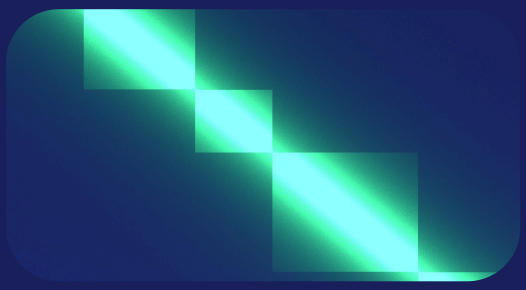
Long-read and full-length isoform sequencing on PacBio HiFi (Revio, Sequel II) and Oxford Nanopore platforms

Small RNA-seq (miRNA)

Small RNA and miRNA sequencing as a standard NGS application

Targeted RNA panels

Targeted RNA panel sequencing on NGS platforms



CATEGORY 04

Epigenomics

5 CAPABILITIES

ChIP-seq

ChIP-seq for genome-wide protein–DNA binding site mapping

CUT&RUN / CUT&Tag

CUT&RUN and CUT&Tag chromatin profiling using established kit chemistries

ATAC-seq

ATAC-seq for genome-wide chromatin accessibility profiling

Bisulfite sequencing / WGBS

WGBS and RRBS methylation sequencing for base-resolution DNA methylation mapping

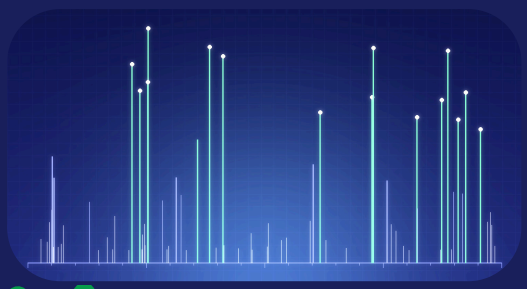
Hi-C / chromosome conformation capture

Proximity-ligation sequencing to map genome-wide chromatin contacts, for 3D genome architecture and for scaffolding genome assemblies

CATEGORY 05

Proteomics

10 CAPABILITIES



Bottom-up (shotgun) proteomics

LC-MS/MS shotgun proteomics on Orbitrap Exploris and Q Exactive-class instruments for protein identification and quantification

Targeted proteomics (MRM/SRM/PRM)

Targeted MRM, SRM and PRM proteomics for quantitative protein panels

Top-down / middle-down proteomics

Intact-protein and middle-down mass spectrometry for proteoform-level characterisation, resolving PTM combinations on the intact molecule rather than inferring them from peptides

PTM-focused proteomics (phospho/glyco/ubiquitin/acetyl)

PTM-enrichment proteomics covering phospho, glyco, ubiquitin and acetyl modifications

Single-cell proteomics

Nanoscale sample preparation coupled to high-sensitivity LC-MS/MS, quantifying proteomes from individual cells

Proximity labeling (BioID/APEX)

Enzyme-catalysed proximity biotinylation from a bait protein, followed by streptavidin enrichment and LC-MS/MS, to map the proximal interactome in living cells

Crosslinking-MS (XL-MS)

Chemical crosslinking followed by LC-MS/MS identification of crosslinked peptide pairs, giving residue-level distance restraints on protein complex architecture

HDX-MS

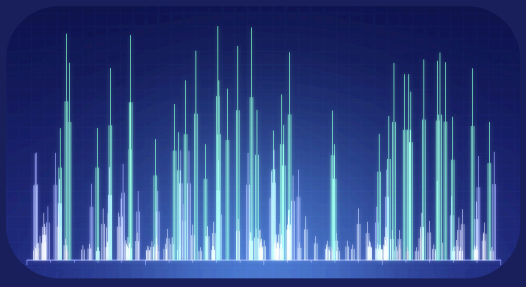
Hydrogen–deuterium exchange with LC-MS readout, mapping backbone solvent accessibility for conformational analysis and epitope-level characterisation

Chemical proteomics / ABPP

Activity-based probe labelling with LC-MS/MS readout, reporting enzyme activity and target engagement across the native proteome, including competition formats for engagement confirmation

Metaproteomics

LC-MS/MS profiling of microbial community samples, with taxonomic and functional assignment of the identified proteins



CATEGORY 06

Metabolomics

4 CAPABILITIES

LC-MS based metabolomics (untargeted/targeted)

Untargeted and targeted LC-MS metabolomics with feature detection and quantification

GC-MS based metabolomics

GC-MS metabolomics for volatile and derivatised metabolite profiling

NMR-based metabolomics

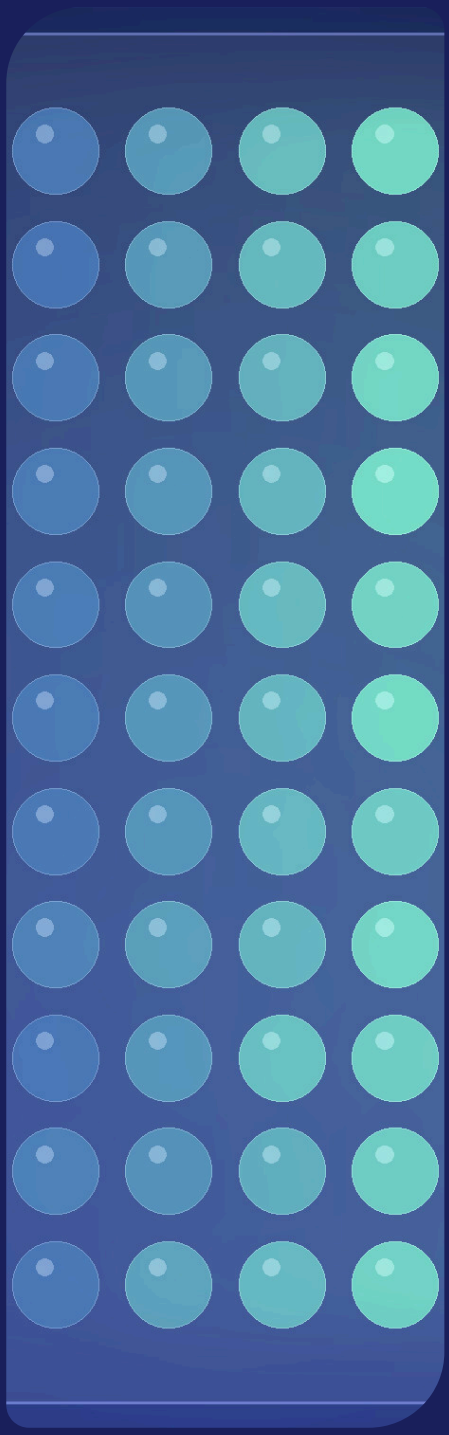
NMR-based metabolomics on high-field instrumentation (600 MHz and above)

Fluxomics

Stable-isotope (13C) labelling with LC-MS/GC-MS readout and metabolic flux modelling, resolving pathway-level flux distributions

SECTION 02

Wet-lab validation

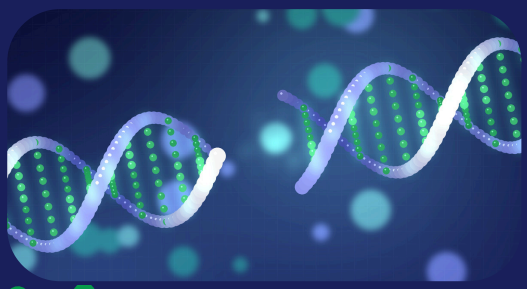


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CATEGORY 07

Genetic Engineering

3 CAPABILITIES



CRISPR knock-out/knock-in gene editing

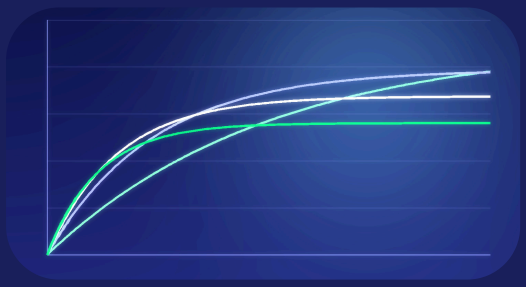
CRISPR knock-out and knock-in editing at 50-85% editing efficiency, with sequence-verified clones in 3-5 weeks and in-house off-target panels

Genome-wide / boutique CRISPR & RNAi screens

Pooled CRISPR and RNAi screening with custom and boutique library designs

Knockout/knock-in phenotypic characterisation

Replicate-level phenotypic characterisation of engineered lines, with automated imaging readouts



CATEGORY 08

Biochemical Screening

6 CAPABILITIES

High-throughput biochemical screening (HTS)

Automated 384-well high-throughput screening at throughputs of tens of thousands of wells per day, with compound library access through partner networks

Enzyme inhibition assays (IC50/Ki)

Enzyme inhibition profiling reporting IC50 and Ki with dose-response and fit-quality metrics

Kinase activity assays

Kinase activity and inhibition assays run against in-house panels of tens to ~100 kinases

Protease activity assays

Fluorogenic and colorimetric protease activity assays

ATPase / GTPase activity assays

ATPase and GTPase activity assays in standard biochemical formats

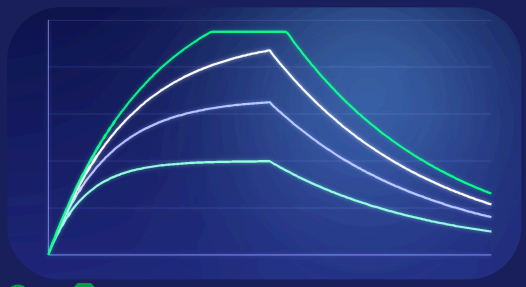
Fragment-based screening (NMR)

Ligand-observed NMR screening (STD, WaterLOGSY) of curated fragment libraries, detecting weak binders for fragment-based lead discovery

CATEGORY 09

Binding & Interaction

6 CAPABILITIES



SPR / BLI / MST (binding kinetics)

Binding kinetics and affinity determination by SPR and BLI on Biacore and Octet instrumentation

ITC (isothermal titration calorimetry)

Isothermal titration calorimetry for label-free binding affinity and thermodynamics

TR-FRET assays

TR-FRET homogeneous proximity assays in kit-based formats with plate-reader detection

AlphaScreen / AlphaLISA proximity assays

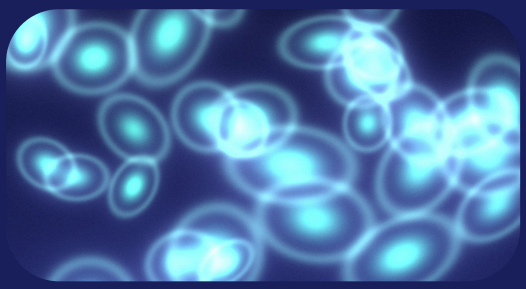
AlphaScreen and AlphaLISA proximity assays using standardised kits and plate-reader detection

Protein–protein interaction mapping

Protein–protein interaction mapping by targeted co-IP and pulldown

Thermal shift / DSF for hit-binding confirmation

Thermal shift and nanoDSF hit-binding confirmation on Prometheus and real-time PCR-based instrumentation



CATEGORY 10

Cell-Based Assays

6 CAPABILITIES

Phenotypic cell-based screening

Phenotypic cell-based screening at hundreds to low thousands of compounds per week

Cell-based functional / receptor assays

FLIPR-class kinetic functional assays for cell-based receptor pharmacology

Reporter gene assays

Reporter gene assay panels for transcriptional and pathway readouts

Cell viability / cytotoxicity assays

Kit-based cell viability and cytotoxicity assays

Apoptosis assays

Flow cytometry-based apoptosis assays

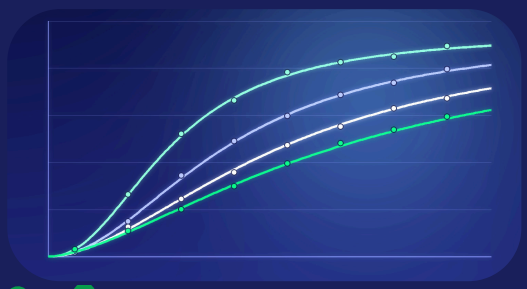
Flow cytometry and immunophenotyping

Cell-population and marker-expression analysis with defined panels and gating strategies

CATEGORY 11

Pharmacology

6 CAPABILITIES



Dose-response / EC50 / IC50 determination

Dose-response curve fitting with EC50 and IC50 determination and replicate-level reporting

Functional agonist/antagonist assays

Functional agonist and antagonist pharmacology assays

Off-target selectivity panels

Off-target selectivity panels, configured per programme

In vivo PK/PD, xenograft efficacy studies

In vivo PK/PD and xenograft efficacy studies in regulated animal-facility settings

Tumour and tissue pharmacodynamics

Tissue-based target-engagement and response assays aligned with pharmacology studies

Quantitative bioanalysis

Small- and large-molecule measurement in blood, plasma, urine, tissue, and other selected biological matrices



CATEGORY 12

Safety Pharmacology & Genotoxicity

2 CAPABILITIES

Electrophysiology / patch clamp (hERG, ion channels)

Automated patch-clamp electrophysiology for hERG and ion-channel liability assessment on multi-well parallel-recording platforms

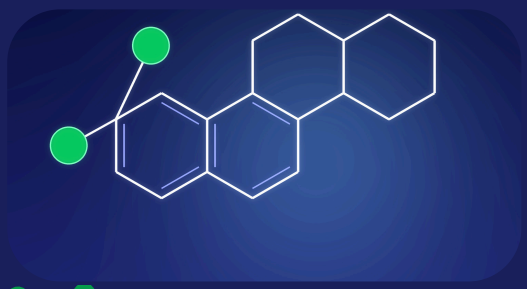
Genotoxicity screening (Ames test)

Ames bacterial reverse-mutation genotoxicity screening, run to OECD test guidelines

CATEGORY 13

ADMET

7 CAPABILITIES



ADMET - solubility

Kinetic and thermodynamic solubility determination as part of core DMPK profiling

ADMET - permeability (PAMPA/Caco-2)

PAMPA and Caco-2 permeability assays for passive and cell-based permeability

ADMET - metabolic stability

Microsomal and hepatocyte metabolic stability assays

ADMET - CYP450 inhibition/induction

CYP450 inhibition and induction panels

ADMET – plasma protein binding

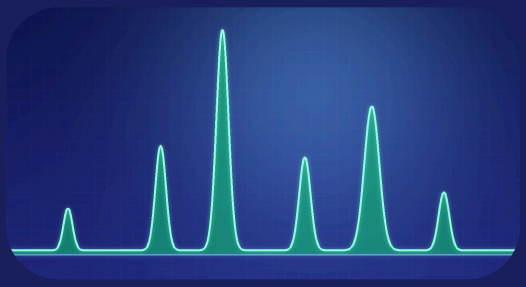
Plasma protein binding determination

ADMET – plasma/blood stability

Plasma and blood stability assays

ADMET – efflux transporter assays

Efflux transporter assays in transfected cell lines



CATEGORY 14

Antibody / Protein Developability

10 CAPABILITIES

Aggregation propensity (SEC-HPLC)

Aggregate and fragment quantification by SEC-HPLC

Thermal stability (Tm) – proteins/antibodies

Thermal stability (Tm) determination for proteins and antibodies by nanoDSF

Hydrophobicity (HIC-HPLC)

Surface hydrophobicity assessment by HIC-HPLC

Charge variant analysis (IEX)

Charge-variant analysis by IEX and icIEF

Polyreactivity (ELISA)

ELISA-based polyreactivity panels for antibody developability

Viscosity assessment

Viscosity and formulation rheology assessment for biologics formulations

Forced degradation / stress stability studies

ICH-aligned forced degradation and stress stability studies

Epitope binning, FcRn binding

SPR-based epitope binning and FcRn binding assessment for antibody developability

Fc effector function assays (ADCC/CDC/ADCP)

Reporter-based Fc effector function assays covering ADCC, CDC and ADCP

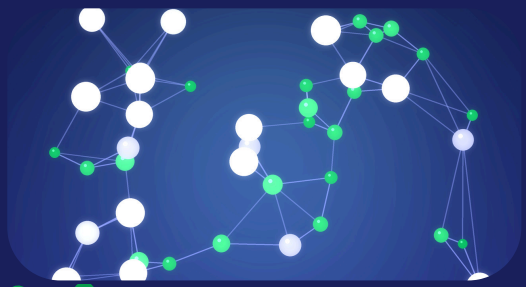
In vitro immunogenicity (T-cell assays)

In vitro immunogenicity assessment by T-cell assays on HLA-diverse donor PBMC banks, with MAPPs peptide-presentation workflows

CATEGORY 15

Protein Engineering

2 CAPABILITIES

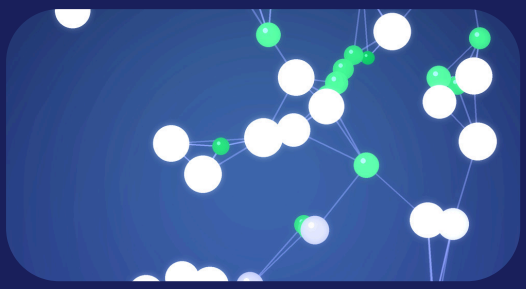


Protein engineering stability assays

Comparative variant-stability screening for engineered proteins

Directed evolution / library screening

Directed evolution and display-library screening



CATEGORY 16

Protein Expression

1 CAPABILITY

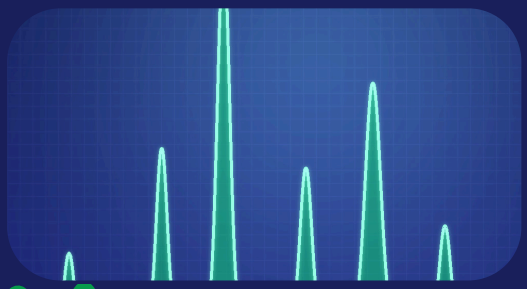
Recombinant protein expression (E. coli / CHO)

Recombinant protein expression in E. coli and CHO systems, with sequence-verified, GMP-track clone banking

CATEGORY 17

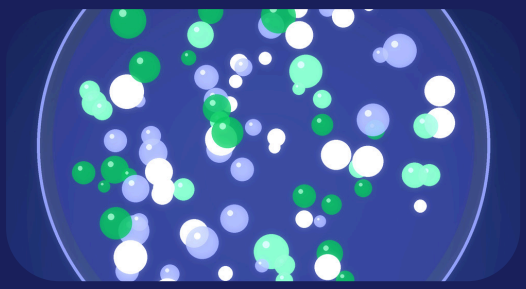
Chemistry

1 CAPABILITY



Medicinal/peptide chemistry synthesis

Medicinal and peptide chemistry synthesis of small molecules and peptides



CATEGORY 18

Microbiology

3 CAPABILITIES

Antimicrobial susceptibility testing (culture/AST)

CLSI-aligned antimicrobial susceptibility testing by culture and AST

Organism identification (MALDI-TOF)

Organism identification by MALDI-TOF mass spectrometry

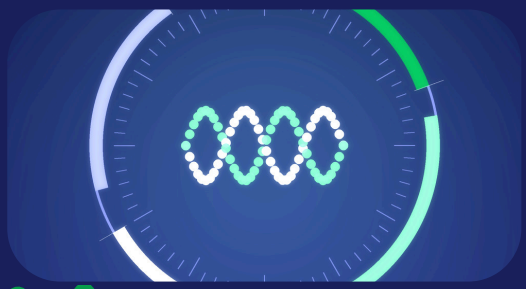
Targeted qPCR resistance-gene panels

qPCR panels for targeted resistance-gene detection on real-time PCR platforms

CATEGORY 19

Synthetic Biology

1 CAPABILITY



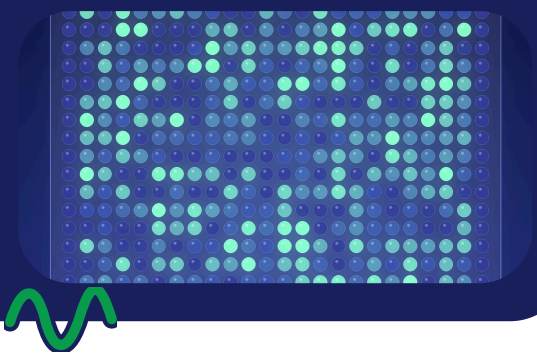
Metabolic pathway engineering / product-titer assays

Metabolic pathway engineering with fermentation process development and product-titer assays at in-house bioreactor scale

SECTION 03

Glossary

58 TERMS



A

- **ABPP** — Activity-based protein profiling — chemical-proteomics approach using reactive probes to report enzyme activity in complex lysates.
- **ADC** — Apparent diffusion coefficient, a quantitative map derived from diffusion-weighted MRI
- **ADCC** — Antibody-dependent cellular cytotoxicity — effector-function assay measuring immune-cell killing of antibody-coated target cells.
- **ADMET** — Absorption, distribution, metabolism, excretion and toxicity — the drug-disposition and safety assay package.
- **ATAC-seq** — Assay for transposase-accessible chromatin with sequencing — maps open chromatin genome-wide.

B

- **BLI** — Bio-layer interferometry — label-free binding-kinetics measurement on a fibre-optic biosensor.
- **BOLD** — Blood-oxygen-level-dependent contrast, the signal underlying functional MRI

C

- **Caco-2** — Human intestinal epithelial cell line used as a cell-based permeability and efflux model.
- **ChIP-seq** — Chromatin immunoprecipitation sequencing — maps protein–DNA binding sites genome-wide.

CRISPR	Clustered regularly interspaced short palindromic repeats — nuclease-guided genome-editing system.
CT	Computed tomography
CUT&RUN / CUT&Tag	Nuclease- or transposase-tethered chromatin profiling methods; lower cell input than ChIP-seq.
CYP450	Cytochrome P450 — the enzyme family dominating hepatic drug metabolism; inhibition and induction are core ADMET endpoints.

D

DEXA	Dual-energy X-ray absorptiometry
DICOM	Digital Imaging and Communications in Medicine, the standard format for clinical image data
DMPK	Drug metabolism and pharmacokinetics.
DSF	Differential scanning fluorimetry (thermal shift) — measures ligand-induced protein thermal stabilisation.
DTI	Diffusion tensor imaging

E

EC50 / IC50	Concentration giving half-maximal effect or half-maximal inhibition.
ECG	Electrocardiogram
ELISA	Enzyme-linked immunosorbent assay.

F

FcRn	Neonatal Fc receptor — binding to it governs antibody serum half-life; a developability endpoint.
FDG	Fluorodeoxyglucose, the most widely used PET tracer
FLAIR	Fluid-attenuated inversion recovery, an MRI sequence suppressing free-fluid signal

G

- **GC-MS** — Gas chromatography–mass spectrometry.

H

- **HDX-MS** — Hydrogen–deuterium exchange mass spectrometry — maps conformation and epitopes via backbone amide exchange.
- **hERG** — Human ether-à-go-go-related gene potassium channel — the cardiac safety liability screened by patch clamp.
- **Hi-C** — Chromosome-conformation capture sequencing — 3D genome contact mapping, also used to scaffold genome assemblies.
- **HIC-HPLC** — Hydrophobic-interaction HPLC — reports surface hydrophobicity of a protein therapeutic.
- **HLA** — Human leukocyte antigen — the MHC locus; relevant to immunogenicity and peptide-presentation assays.
- **HPLC** — High-performance liquid chromatography.
- **HTS** — High-throughput screening.

I

- **IEX** — Ion-exchange chromatography — charge-variant analysis of protein therapeutics.
- **ITC** — Isothermal titration calorimetry — label-free measurement of binding affinity and thermodynamics.

K

- **Ki** — Inhibition constant — concentration-independent measure of inhibitor potency.

L

- **LC-MS/MS** — Liquid chromatography with tandem mass spectrometry.

M

- **MALDI-TOF** — Matrix-assisted laser desorption/ionisation time-of-flight mass spectrometry.
- **MAPPs** — MHC-associated peptide proteomics — identifies presented peptides to assess immunogenicity risk.
- **MRI** — Magnetic resonance imaging
- **MRM / PRM** — Multiple- and parallel-reaction monitoring — targeted, quantitative mass-spectrometry readouts.
- **MRS** — Magnetic resonance spectroscopy
- **MST** — Microscale thermophoresis — binding affinity from the motion of molecules in a temperature gradient.

N

- **NGS** — Next-generation sequencing.
- **NMR** — Nuclear magnetic resonance spectroscopy.

O

- **OCT** — Optical coherence tomography

P

- **PAMPA** — Parallel artificial membrane permeability assay — cell-free passive-permeability screen.
- **PBMC** — Peripheral blood mononuclear cells.
- **PET** — Positron emission tomography
- **PTM** — Post-translational modification.

R

- **RNA-seq** — RNA sequencing; scRNA-seq denotes the single-cell form.

- **RNAi** — RNA interference — sequence-specific knockdown, used for genetic screens.

S

- **SEC-HPLC** — Size-exclusion HPLC — aggregate and fragment quantification for protein therapeutics.
- **SMRT / HiFi** — Single-molecule real-time long-read sequencing and its high-accuracy consensus reads.
- **SPECT** — Single-photon emission computed tomography
- **SPR** — Surface plasmon resonance — label-free, real-time binding kinetics and affinity.

T

- **TR-FRET** — Time-resolved fluorescence resonance energy transfer — proximity-based homogeneous assay format.

W

- **WGBS** — Whole-genome bisulfite sequencing — base-resolution DNA methylation mapping.

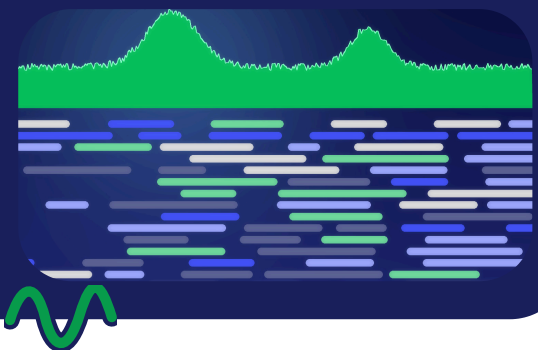
X

- **XL-MS** — Crosslinking mass spectrometry — maps residue-level contacts in protein complexes.

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